

%L1.c. Stability Analysis

% Problem 1

% Determine the stability of the negative unity feedback system

% whose open loop transfer function is given by

% $G(s)=1.25/(s(s+1)(0.5s+1))$

% Numerator coefficient

a=[1.25];

% Denominator coefficients for $s(s+1)$

b=[1 1 0];

% Denominator coefficients for $(0.5s+1)$

c=[0.5 1];

% Overall denominator

d=conv(b,c);

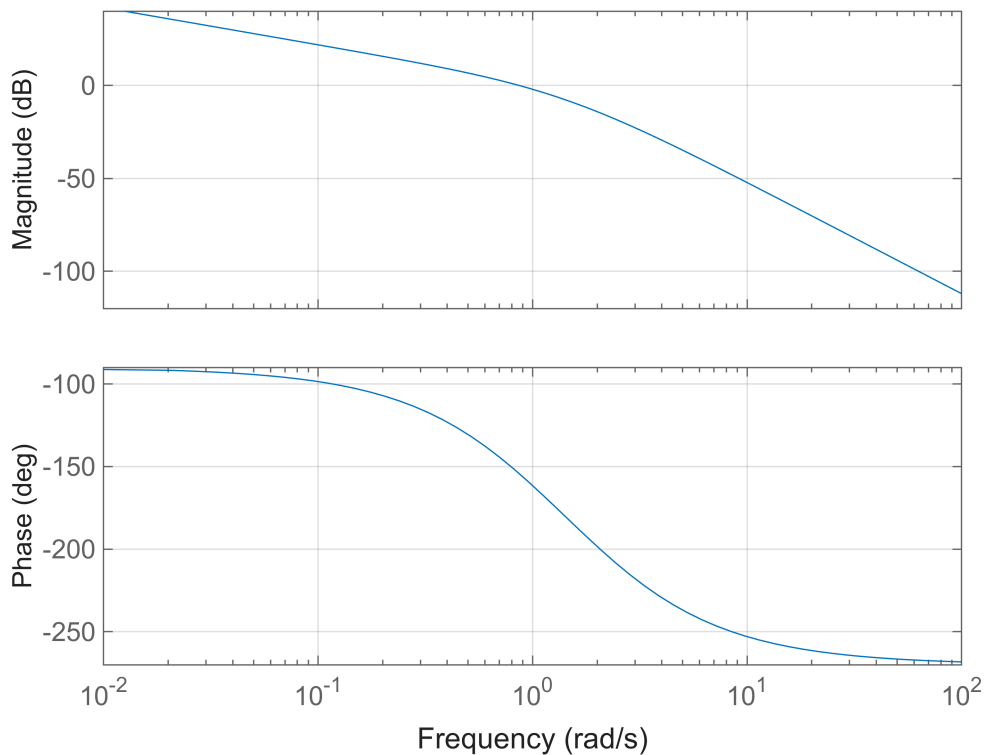
% Bode plot

figure(1)

bode(a,d)

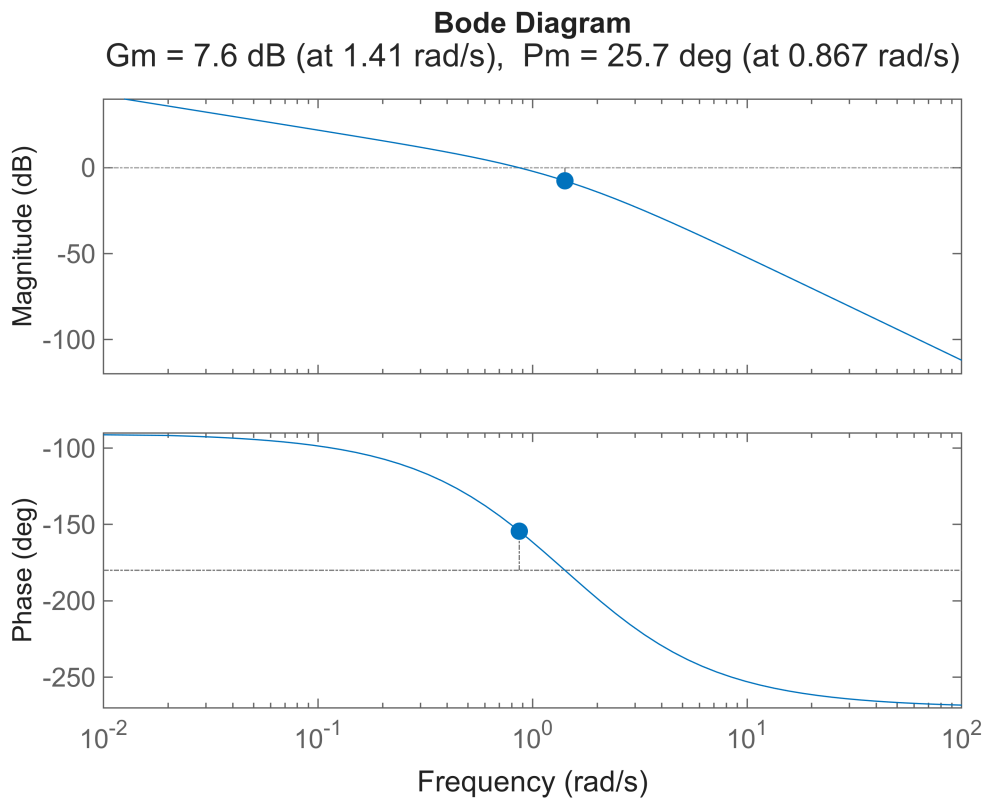
grid on

Bode Diagram

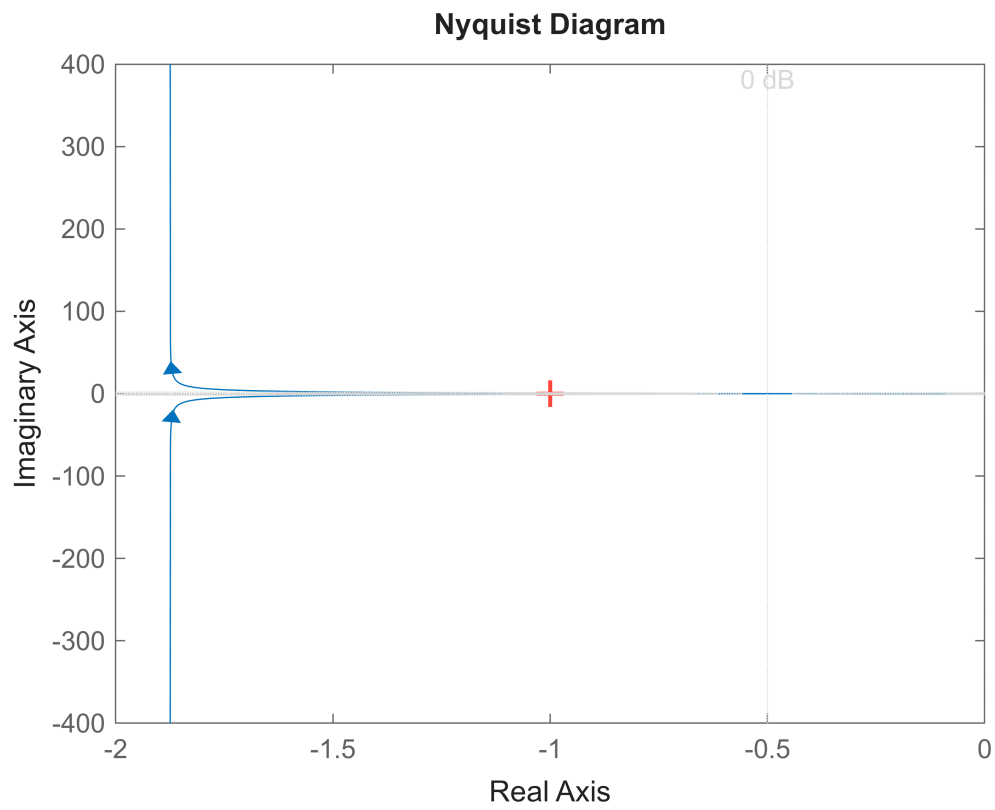


% Gain margin and phase margin

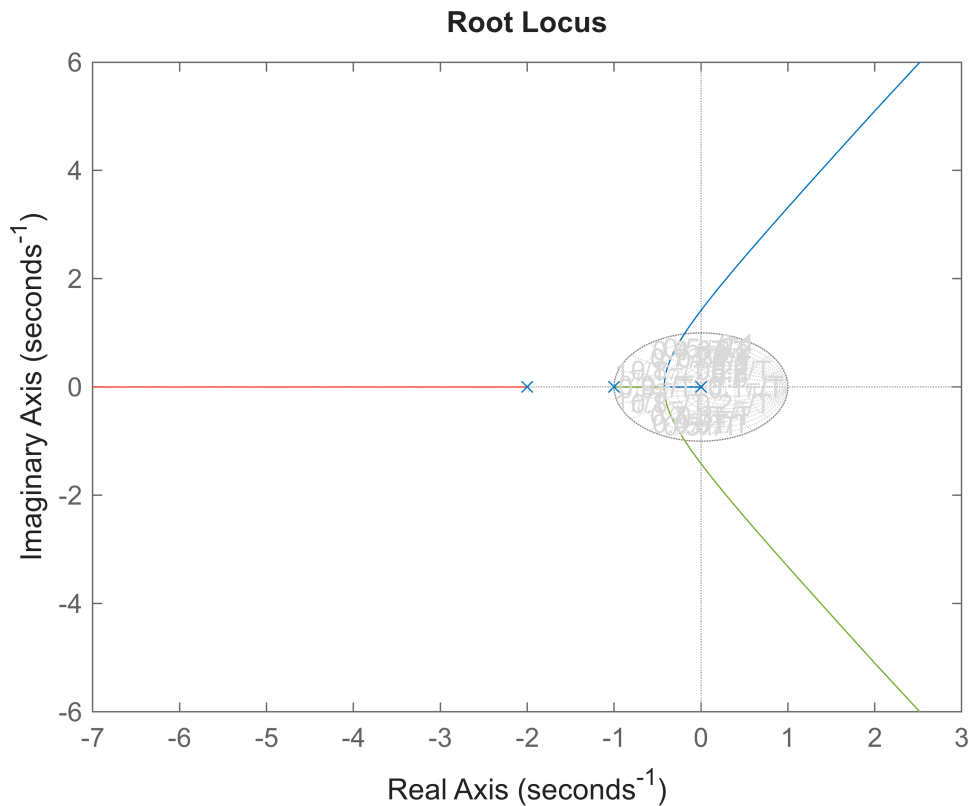
margin(a,d)



```
% Nyquist plot  
figure(2)  
nyquist(a,d)  
grid on
```



```
% Root locus  
figure(3)  
rlocus(a,d)  
grid on
```



%L1.c. Stability Analysis

% Problem 2

% Sketch the root locus for the system

% $G(s)H(s) = K(s^2 - 4s + 20) / ((s+2)(s+4))$

% Find:

% a) Point and gain where locus crosses $\zeta = 0.45$ line

% b) Point and gain where locus crosses imaginary axis

% c) Breakaway point on real axis

% d) Range of K for stability

% Numerator coefficients

numgh=[1 -4 20];

% Denominator coefficients

dengh=poly([-2 -4]);

% Transfer function

GH=tf(numgh,dengh);

% Root locus with damping ratio grid

figure(1)

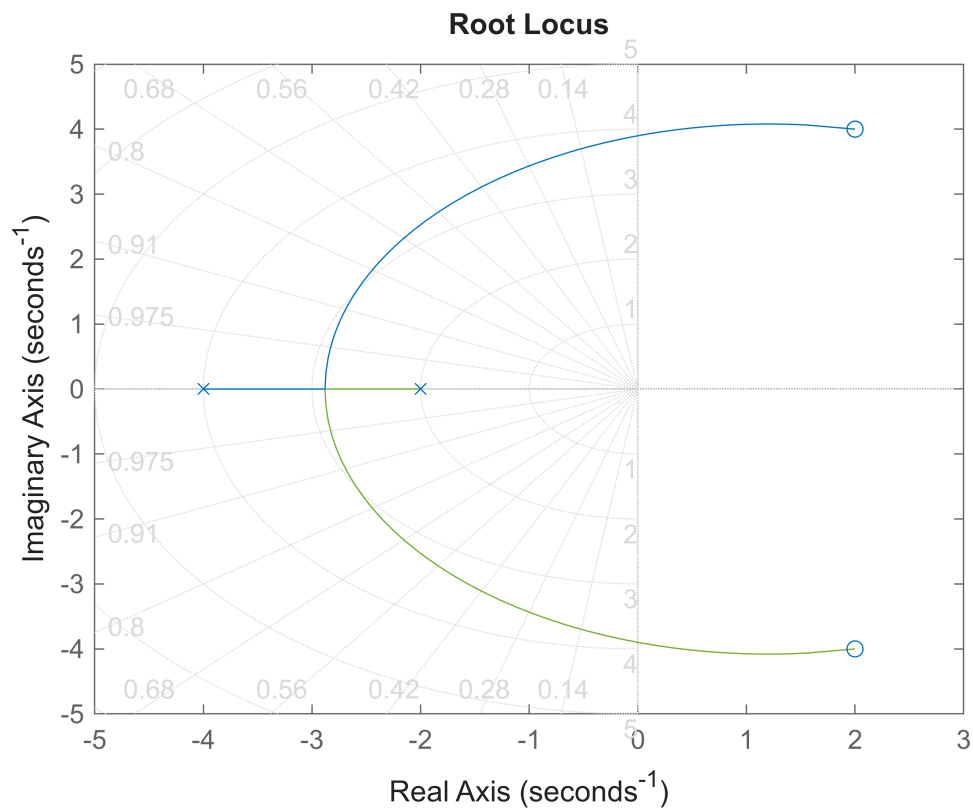
rlocus(GH)

z=0.2:0.05:0.5;

wn=0:1:10;

sgrid(z,wn)

grid on



```
% Root locus for zeta=0.45
```

```
figure(2)
```

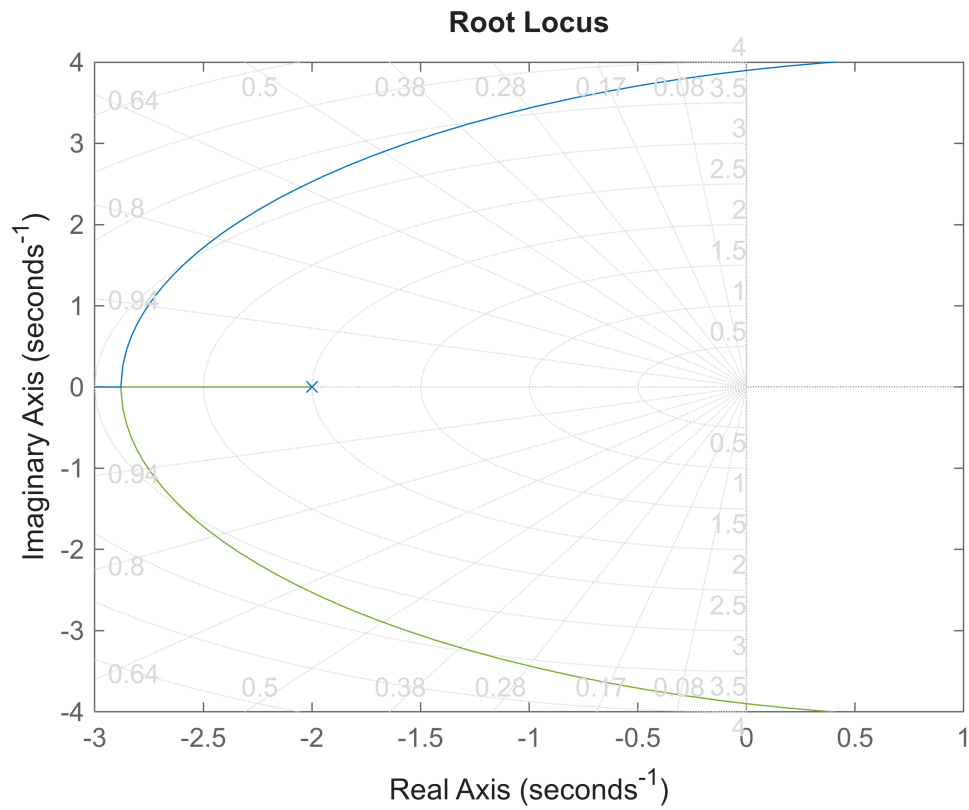
```
rlocus(GH)
```

```
axis([-3 1 -4 4])
```

```
z=0.45;
```

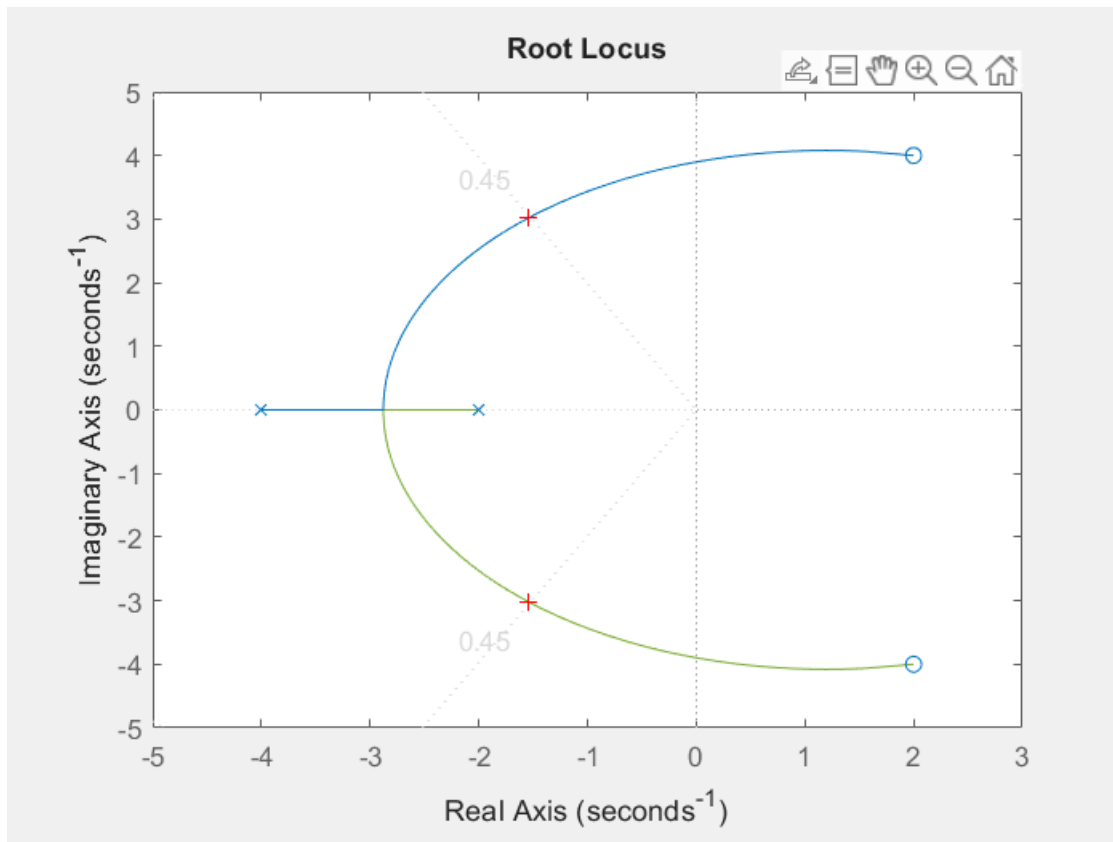
```
wn=0;
```

```
sgrid(z,wn)
```



```
% Select point where locus crosses zeta=0.45 line
[k,p]=rlocfind(GH)
```

Select a point in the graphics window



```
selected_point =
-1.5157 + 2.9975i
k =
0.4129
p = 2x1 complex
-1.5388 + 3.0231i
-1.5388 - 3.0231i
```

```
% Closed-loop poles
for i=1:length(p)
    p(i)
end
```

```
ans =
-1.5388 + 3.0231i
ans =
-1.5388 - 3.0231i
```

%L1.c. Stability Analysis

```
% Exercise 1
% Draw Bode diagram for the transfer function
%  $G(s)=e^{(-2s)}/(s(s+1))$ 
% Determine the stability
```

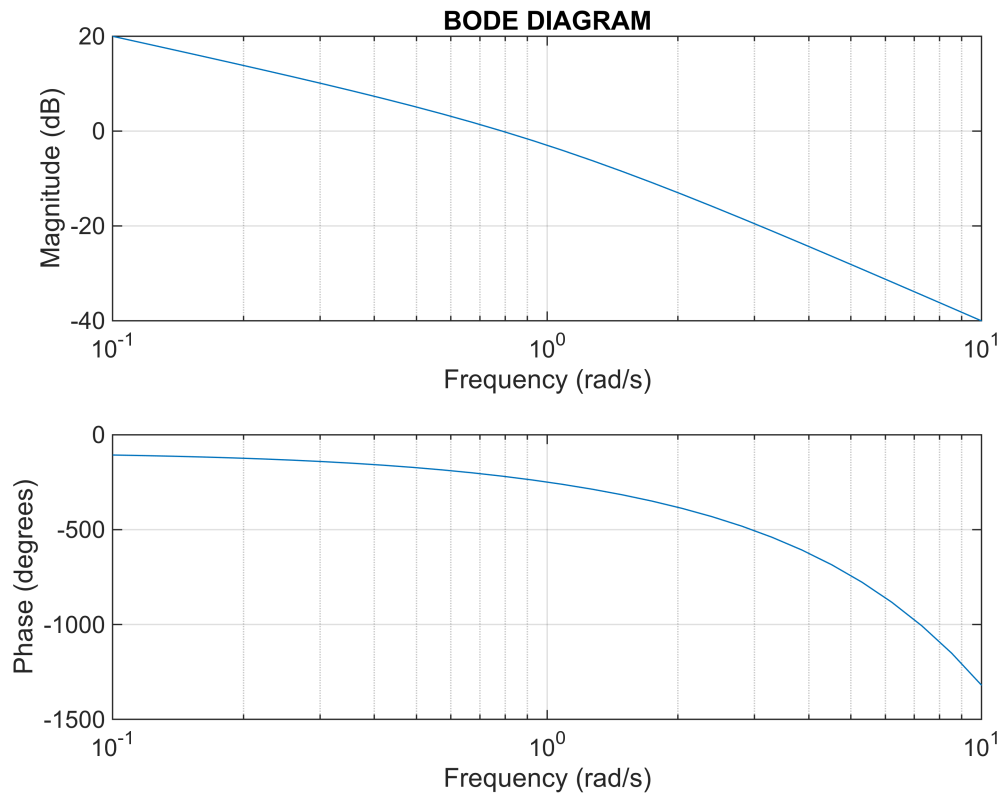
```
% Frequency range
a=-1;
b=1;
nn=30;
```

```

w=logspace(a,b,nn);

% Transfer function coefficients for 1/(s(s+1))
Gnum=[1];
Gden=[1 1 0];
% Magnitude and phase response
[mag,phase,w]=bode(Gnum,Gden,w);
% Convert magnitude to dB
db=20*log10(squeeze(mag));
% Convert phase to vector
phase=squeeze(phase);
% Additional phase due to delay e^(-2s)
phased1=-2*57.296*w;
% Total phase
phase=phase+phased1;
% Bode magnitude plot
subplot(2,1,1)
semilogx(w,db)
title('BODE DIAGRAM')
xlabel('Frequency (rad/s)')
ylabel('Magnitude (dB)')
grid on
% Bode phase plot
subplot(2,1,2)
semilogx(w,phase)
xlabel('Frequency (rad/s)')
ylabel('Phase (degrees)')
grid on

```

%L1.c. Stability Analysis

% Exercise 2

% Obtain the polar and Nyquist plot of the system having

% $G(s)H(s)=k/(s(s+0.5)(s+2))$

% Also determine gain crossover frequency, phase margin,

% phase crossover frequency and gain margin

% Open loop gain

k=input('Enter the value of open loop gain k = ');

% Open loop poles

pol=input('Enter three open loop poles [p1 p2 p3] = ');

% Numerator coefficient

n=k;

% Denominator coefficients

d=poly(pol);

% Open loop transfer function

sol=tf(n,d);

% Nyquist data

[re,im,w]=nyquist(sol);

% Magnitude

mag=sqrt(re(:).^2+im(:).^2);

% Magnitude in dB

magdb=20*log10(mag);

% Phase in degrees

```

phd=(180/pi)*atan2(im(:),re(:));
% Display frequency response data
c=[w magdb phd];
disp(c)
% Polar plot
figure(1)
plot(re(:),im(:))
title('Polar Plot')
xlabel('Real Axis')
ylabel('Imaginary Axis')
grid on
% Nyquist plot
figure(2)
plot(re(:),im(:),re(:),-im(:))
title('Nyquist Plot')
xlabel('Real Axis')
ylabel('Imaginary Axis')
grid on
% Unit circle
[x,y]=scircle1(0,0,1);
hold on
plot(x,y)
% Nyquist plot using MATLAB command
figure(3)
nyquist(n,d)
grid on
% Gain margin, phase margin and crossover frequencies
figure(4)
margin(n,d)
grid on
% Open loop transfer function
oltf=tf(n,d);
% Closed loop transfer function
cltf=feedback(oltf,1);
% Closed loop numerator and denominator
[ncl,dcl]=tfdata(cltf,'v');
% Closed loop poles
clpoles=pole(cltf)

```

%L1.c. Stability Analysis

```

% Exercise 3(i)
% Obtain the Nyquist plot and determine the stability
%  $G(s)=15(s+5)/(s(s+2)(s^2+6s+15))$ 

% Numerator coefficients
num=[15 75];
% Denominator coefficients
den=conv([1 0],[1 2]);

```

```

den=conv(den,[1 6 15]);
% Open loop transfer function
G=tf(num,den);
% Nyquist plot
figure(1)
nyquist(G)
grid on
% Gain margin and phase margin
figure(2)
margin(G)
grid on
% Closed loop transfer function
CL=feedback(G,1);
% Closed loop poles
clpoles=pole(CL)

```

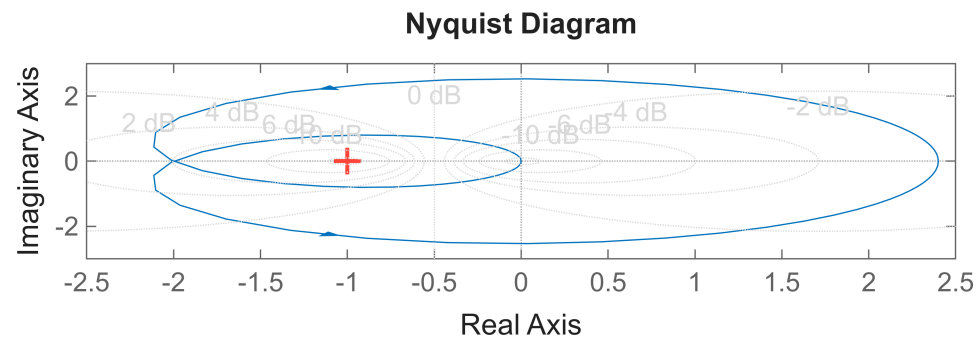
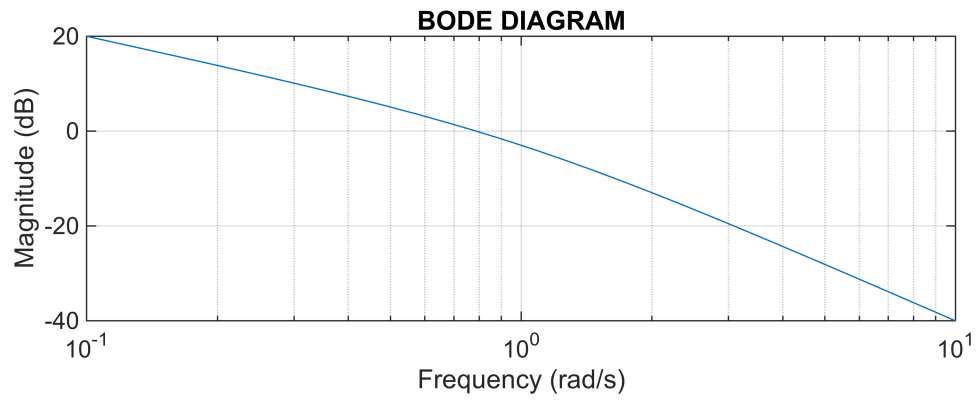
%L1.c. Stability Analysis

```

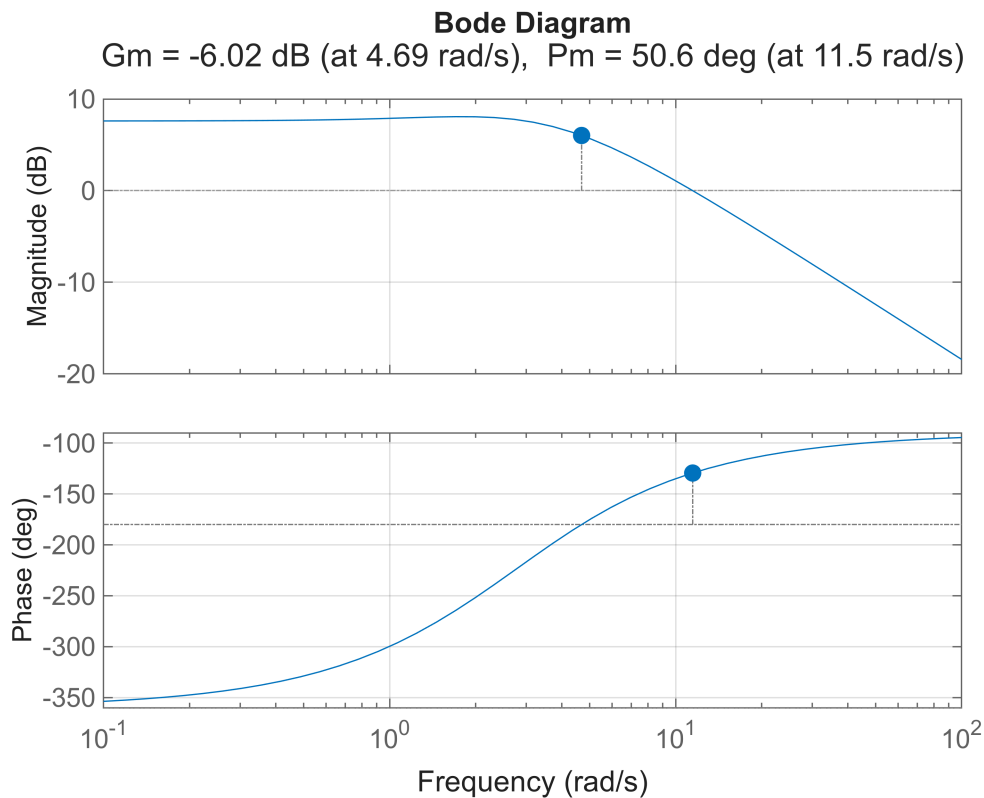
% Exercise 3(ii)
% Obtain the Nyquist plot and determine the stability
%  $G(s)=12(s+2)/(s^2-6s+10)$ 

% Numerator coefficients
num=[12 24];
% Denominator coefficients
den=[1 -6 10];
% Open loop transfer function
G=tf(num,den);
% Nyquist plot
figure(1)
nyquist(G)
grid on

```



```
% Gain margin and phase margin  
figure(2)  
margin(G)  
grid on
```



```
% Closed loop transfer function
```

```
CL=feedback(G,1);
```

```
% Closed loop poles
```

```
clpoles=pole(CL)
```

```
clpoles = 2×1 complex
```

```
 -3.0000 + 5.0000i
```

```
 -3.0000 - 5.0000i
```

```
%L1.c. Stability Analysis
```

```
% Exercise 4(i)
```

```
% Obtain the root locus plot for
```

```
%  $G(s)=K/(s(s+4)(s+8))$ 
```

```
% Find the stable range of K and break-away points
```

```
% Numerator coefficients
```

```
num=[1];
```

```
% Denominator coefficients
```

```
den=conv([1 0],[1 4]);
```

```
den=conv(den,[1 8]);
```

```
% Transfer function
```

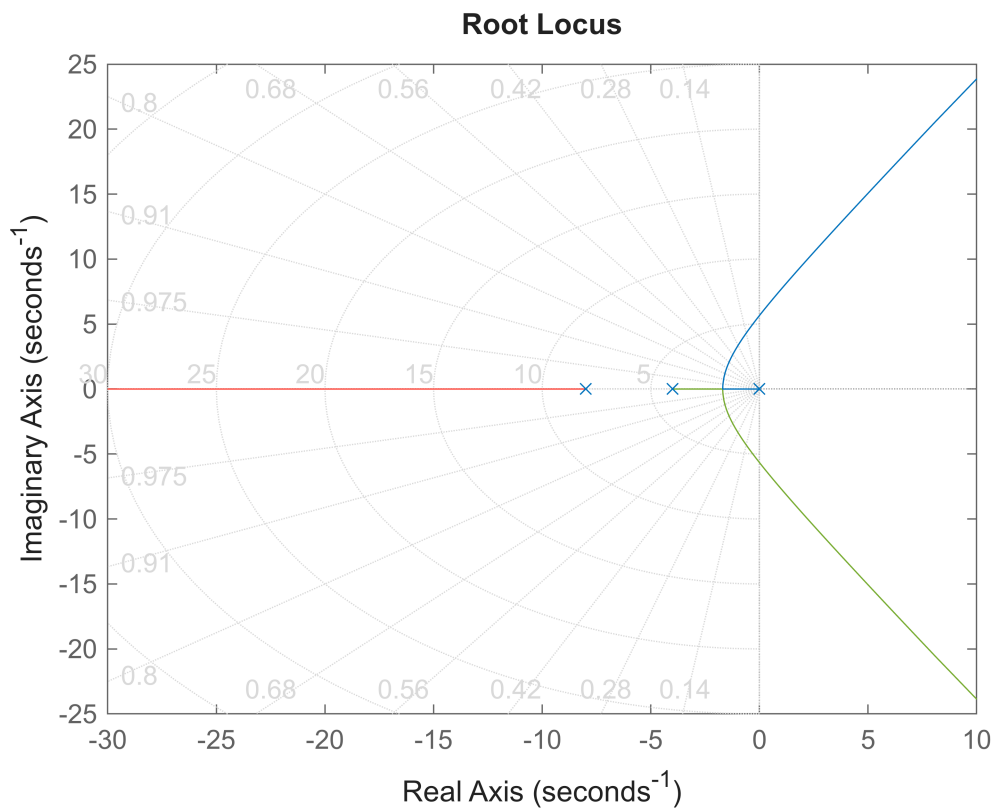
```
G=tf(num,den);
```

```
% Root locus
```

```
figure(1)
```

```
rlocus(G)
```

grid on

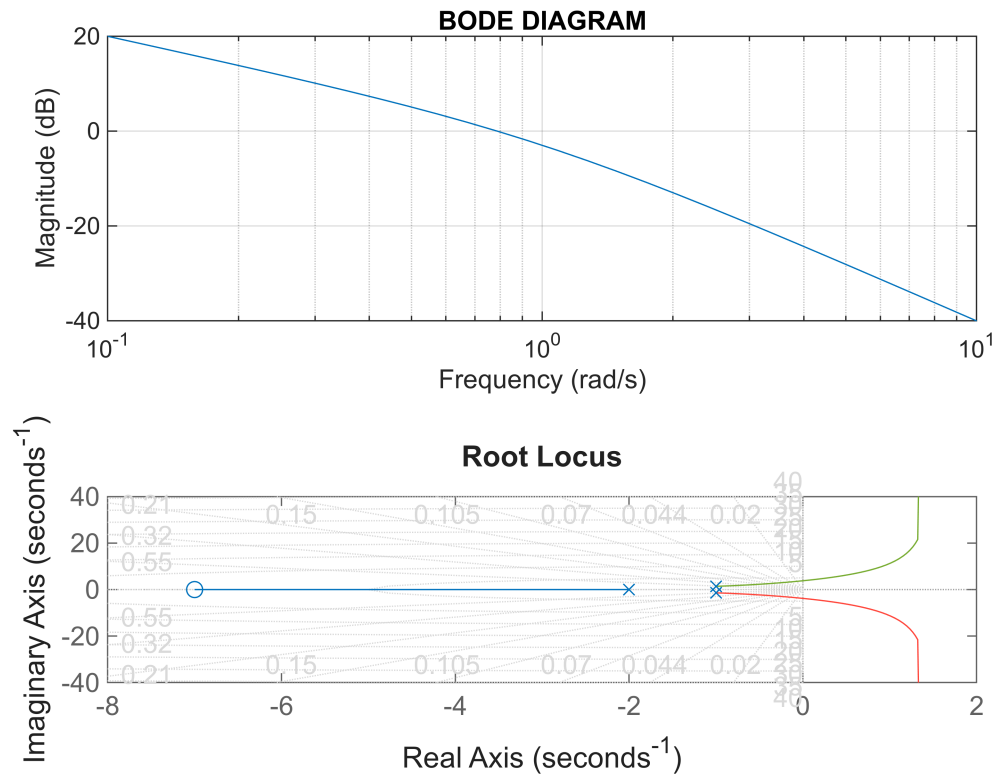


```
% Gain and pole locations from root locus  
[k,p]=rlocfind(G)
```

%L1.c. Stability Analysis

```
% Exercise 4(ii)  
% Obtain the root locus plot for  
%  $G(s)=K(s+7)/((s+2)(s^2+2s+3))$   
% Find the stable range of K and break-away points
```

```
% Numerator coefficients  
num=[1 7];  
% Denominator coefficients  
den=conv([1 2],[1 2 3]);  
% Transfer function  
G=tf(num,den);  
% Root locus  
figure(1)  
rlocus(G)  
grid on
```



```
% Gain and pole locations from root locus
[k,p]=rlocfind(G)
```